

Teaching Expertise – Prof. Dr. Boris Vertman

Supervised Ph.D's and Postdocs

- Postdocs: **Dr. Matheus Santos** (2025-2026)
funded by FAPESP, Sao Paolo, Brazil
- Dr. Mannaim Vitti** (2024)
funded by FAPESP, Sao Paolo, Brazil
- Dr. Tobias Marxen** (2017-2024)
funded by Geometry at Infinity SPP (DFG)
- Dr. Jorgen Olsen Lye** (2021-2023)
funded by Geometry at Infinity SPP (DFG)
- current Ph.D's: **Paul Hafemann** (starting in October 2025)
Cartan geometric formulation of gravity
- finished Ph.D's **Alvaro Sanchez** (graduation September 2025)
Index theory on manifolds with fibred boundaries
- Orville Damaschke** (graduated summer 2023)
Gamma index theory on Lorentzian space-times
- Guiseppe Gentile** (graduated winter 2022)
Mean curvature flow in Lorentzian space-times
- Bruno Caldera** (graduation in September 2021)
Renormalized Yamabe flow on non-compact manifolds
- Mohammad Talebi** (graduated winter 2020)
Spectral geometry for spaces with fibered boundary.
- Marcel Wunderlich** (graduated summer 2019)
Mean curvature flow with conical singularities

Supervised M.Sc. & B.Sc.

- finished M.Sc. **Paul Hafemann** (graduated in August 2023)
Extremals of determinants of discrete Laplacians
- Sina Held** (graduated in September 2021)
Allen-Cahn phase transition on singular spaces
- Alexander Meiners** (graduated in September 2021)
Discrete zeta function and Epstein-Riemann hypothesis
- Martin Dombeck** (graduated in February 2021)
Kirchhoff theorem and the dimer model
- Benedikt Sauer** (graduated in June 2015)
Regularizing infinite sums of determinants
- finished B.Sc. numerous B.Sc. in Bonn, Münster and Oldenburg.

Courses given at the University Oldenburg

lectures	Math. Foundations of Data Science (2024) Analysis I, II, III , (2017-18, 2020-21, 2024-26) Differential geometry , (2020, 2023) Geometric Analysis, ΨDO , (2021) Mathematics in Biology 2 , (2019) Functional Analysis , (2019) Topology I , (2023) Complex Analysis , (2018, 2019, 2022) Partial Differential Equations , (2018) Elementary PDE , (2018, 2020, 2023). Spectral Theory , (2017, 2019)
seminars	Calabi-Yau conjecture on Fano manifolds Wave equations and General relativity Atiyah-Singer index theorem Microlocal analysis & many more.

Courses given at the University Münster

Winter 2016/17	Differential Geometry I , lectures and exercises
Winter 2015/16	Complex Analysis , (Funktionentheorie für Lehramt)
Summer 2015	Fourier series and Fourier integrals , student seminar
Winter 2014/15	Geometric Analysis , lectures and exercises

Courses given at the University Bonn

Summer 2016	Introduction into Kähler Geometry and, Kähler Einstein Metrics , lectures and exercises (held as W ₃ substitute professor at Bonn)
Summer 2014	Geometric Analysis II , lectures and exercises
Winter 2012/13	Analysis III , lectures and exercises
Sommer 2012	Analysis II , lectures and exercises
Winter 2010/11	Microlocal analysis and b-calculus , lectures
Summer 2013	Exercises in Geometry and Topology

Summer 2011	Special functions in mathematical physics , seminar
Winter 2010/11	The Index of Elliptic Operators , seminar
Summer 2009	Topological Quantum Field Theory , seminar